**6.2.10 Examples** **(a)** The exponential function $f(x) := e^x$ has the derivative $f'(x) = e^x$ for all $x \in \mathbb{R}$. Thus $f'(x) > 1$ for $x > 0$, and $f'(x) < 1$ for $x < 0$. From these relationships, we will derive the inequality

$$e^x \geq 1 + x \quad \text{for} \quad x \in \mathbb{R}, \tag{1}$$

with equality occurring if and only if $x = 0$.

If $x = 0$, we have equality with both sides equal to 1. If $x > 0$, we apply the Mean Value Theorem to the function $f$ on the interval $[0, x]$. Then for some $c$ with $0 < c < x$ we have

$$e^x - e^0 = e^c(x - 0).$$

Since $e^0 = 1$ and $e^c > 1$, this becomes $e^x - 1 > x$ so that we have $e^x > 1 + x$ for $x > 0$. A similar argument establishes the same strict inequality for $x < 0$. Thus the inequality (1) holds for all $x$, and equality occurs only if $x = 0$.

**(b)** The function $g(x) := \sin x$ has the derivative $g'(x) = \cos x$ for all $x \in \mathbb{R}$. On the basis of the fact that $-1 \leq \cos x \leq 1$ for all $x \in \mathbb{R}$, we will show that

$$-x \leq \sin x \leq x \quad \text{for all} \quad x \geq 0. \tag{2}$$

Indeed, if we apply the Mean Value Theorem to $g$ on the interval $[0, x]$, where $x > 0$, we obtain

$$\sin x - \sin 0 = (\cos c)(x - 0)$$

for some $c$ between 0 and $x$. Since $\sin 0 = 0$ and $-1 \leq \cos c \leq 1$, we have $-x \leq \sin x \leq x$. Since equality holds at $x = 0$, the inequality (2) is established.

**(c)** (Bernoulli's inequality) If $\alpha > 1$, then

$$(1 + x)^\alpha \geq 1 + \alpha x \quad \text{for all} \quad x > -1, \tag{3}$$

with equality if and only if $x = 0$.

This inequality was established earlier, in Example 2.1.13(c), for positive integer values of $\alpha$ by using Mathematical Induction. We now derive the more general version by employing the Mean Value Theorem.

If $h(x) := (1 + x)^\alpha$ then $h'(x) = \alpha(1 + x)^{\alpha - 1}$ for all $x > -1$. [For rational $\alpha$ this derivative was established in Example 6.1.10(c). The extension to irrational will be discussed in Section 8.3.] If $x > 0$, we infer from the Mean Value Theorem applied to $h$ on the interval $[0, x]$ that there exists $c$ with $0 < c < x$ such that $h(x) - h(0) = h'(c)(x - 0)$. Thus, we have

$$(1 + x)^\alpha - 1 = \alpha(1 + c)^{\alpha - 1}x.$$

Since $c > 0$ and $\alpha - 1 > 0$, it follows that $(1 + c)^{\alpha - 1} > 1$ and hence that $(1 + x)^\alpha > 1 + \alpha x$. If $-1 < x < 0$, a similar use of the Mean Value Theorem on the interval $[x, 0]$ leads to the same strict inequality. Since the case $x = 0$ results in equality, we conclude that (3) is valid for all $x > -1$ with equality if and only if $x = 0$.

**(d)** Let $\alpha$ be a real number satisfying $0 < \alpha < 1$ and let $g(x) = \alpha x - x^\alpha$ for $x \geq 0$. Then $g'(x) = \alpha(1 - x^{\alpha - 1})$, so that $g'(x) < 0$ for $0 < x < 1$ and $g'(x) > 0$ for $x > 1$. Consequently, if $x \geq 0$, then $g(x) \geq g(1)$ and $g(x) = g(1)$ if and only if $x = 1$. Therefore, if $x \geq 0$ and $0 < \alpha < 1$, then we have

$$x^\alpha \leq \alpha x + (1 - \alpha).$$

If $a > 0$ and $b > 0$ and if we let $x = a/b$ and multiply by $b$, we obtain the inequality

$$a^{\alpha}b^{1-\alpha} \leq \alpha a + (1-\alpha)b,$$

where equality holds if and only if $a = b$. □

### The Intermediate Value Property of Derivatives

We conclude this section with an interesting result, often referred to as Darboux's Theorem. It states that if a function $f$ is differentiable at every point of an interval $I$, then the function $f'$ has the Intermediate Value Property. This means that if $f'$ takes on values $A$ and $B$, then it also takes on all values between $A$ and $B$. The reader will recognize this property as one of the important consequences of continuity as established in Theorem 5.3.7. It is remarkable that derivatives, which need not be continuous functions, also possess this property.

**6.2.11 Lemma** *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, let $c \in I$, and assume that $f$ has a derivative at $c$. Then:*

**(a)** *If $f'(c) > 0$, then there is a number $\delta > 0$ such that $f(x) > f(c)$ for $x \in I$ such that $c < x < c + \delta$.*

**(b)** *If $f'(c) < 0$, then there is a number $\delta > 0$ such that $f(x) > f(c)$ for $x \in I$ such that $c - \delta < x < c$.*

***Proof.*** (a) Since

$$\lim_{x \to c} \frac{f(x) - f(c)}{x - c} = f'(c) > 0,$$

it follows from Theorem 4.2.9 that there is a number $\delta > 0$ such that if $x \in I$ and $0 < |x - c| < \delta$, then

$$\frac{f(x) - f(c)}{x - c} > 0.$$

If $x \in I$ also satisfies $x > c$, then we have

$$f(x) - f(c) = (x - c) \cdot \frac{f(x) - f(c)}{x - c} > 0.$$

Hence, if $x \in I$ and $c < x < c + \delta$, then $f(x) > f(c)$.

The proof of (b) is similar. Q.E.D.

**6.2.12 Darboux's Theorem** *If $f$ is differentiable on $I = [a, b]$ and if $k$ is a number between $f'(a)$ and $f'(b)$, then there is at least one point $c$ in $(a, b)$ such that $f'(c) = k$.*

***Proof.*** Suppose that $f'(a) < k < f'(b)$. We define $g$ on $I$ by $g(x) := kx - f(x)$ for $x \in I$. Since $g$ is continuous, it attains a maximum value on $I$. Since $g'(a) = k - f'(a) > 0$, it follows from Lemma 6.2.11(a) that the maximum of $g$ does not occur at $x = a$. Similarly, since $g'(b) = k - f'(b) < 0$, it follows from Lemma 6.2.11(b) that the maximum does not occur at $x = b$. Therefore, $g$ attains its maximum at some $c$ in $(a, b)$. Then from Theorem 6.2.1 we have $0 = g'(c) = k - f'(c)$. Hence, $f'(c) = k$. Q.E.D.

**6.2.13 Example** The function $g$: $[-1, 1] \to \mathbb{R}$ defined by

$$g(x) := \begin{cases} 1 & \text{for } 0 < x \leq 1, \\ 0 & \text{for } x = 0, \\ -1 & \text{for } -1 \leq x < 0, \end{cases}$$

(which is a restriction of the signum function) clearly fails to satisfy the intermediate value property on the interval $[-1, 1]$. Therefore, by Darboux's Theorem, there does not exist a function $f$ such that $f'(x) = g(x)$ for all $x \in [-1, 1]$. In other words, $g$ is *not* the derivative on $[-1, 1]$ of any function. □

## Exercises for Section 6.2

1. For each of the following functions on $\mathbb{R}$ to $\mathbb{R}$, find points of relative extrema, the intervals on which the function is increasing, and those on which it is decreasing:
   (a) $f(x) := x^2 - 3x + 5$, (b) $g(x) := 3x - 4x^2$,
   (c) $h(x) := x^3 - 3x - 4$, (d) $k(x) := x^4 + 2x^2 - 4$.

2. Find the points of relative extrema, the intervals on which the following functions are increasing, and those on which they are decreasing:
   (a) $f(x) := x + 1/x$ for $x \neq 0$, (b) $g(x) := x/(x^2 + 1)$ for $x \in \mathbb{R}$,
   (c) $h(x) := \sqrt{x} - 2\sqrt{x+2}$ for $x > 0$, (d) $k(x) := 2x + 1/x^2$ for $x \neq 0$.

3. Find the points of relative extrema of the following functions on the specified domain:
   (a) $f(x) := |x^2 - 1|$ for $-4 \leq x \leq 4$, (b) $g(x) := 1 - (x-1)^{2/3}$ for $0 \leq x \leq 2$,
   (c) $h(x) := x|x^2 - 12|$ for $-2 \leq x \leq 3$, (d) $k(x) := x(x-8)^{1/3}$ for $0 \leq x \leq 9$.

4. Let $a_1, a_2, \ldots, a_n$ be real numbers and let $f$ be defined on $\mathbb{R}$ by
$$f(x) := \sum_{i=1}^{n} (a_i - x)^2 \quad \text{for} \quad x \in \mathbb{R}.$$
   Find the unique point of relative minimum for $f$.

5. Let $a > b > 0$ and let $n \in \mathbb{N}$ satisfy $n \geq 2$. Prove that $a^{1/n} - b^{1/n} < (a-b)^{1/n}$. [*Hint*: Show that $f(x) := x^{1/n} - (x-1)^{1/n}$ is decreasing for $x \geq 1$, and evaluate $f$ at 1 and $a/b$.]

6. Use the Mean Value Theorem to prove that $|\sin x - \sin y| \leq |x - y|$ for all $x$, $y$ in $\mathbb{R}$.

7. Use the Mean Value Theorem to prove that $(x-1)/x < \ln x < x - 1$ for $x > 1$. [*Hint*: Use the fact that $D \ln x = 1/x$ for $x > 0$.]

8. Let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$ and differentiable in $(a, b)$. Show that if $\lim_{x \to a} f'(x) = A$, then $f'(a)$ exists and equals $A$. [*Hint*: Use the definition of $f'(a)$ and the Mean Value Theorem.]

9. Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) := 2x^4 + x^4 \sin(1/x)$ for $x \neq 0$ and $f(0) := 0$. Show that $f$ has an absolute minimum at $x = 0$, but that its derivative has both positive and negative values in every neighborhood of 0.

10. Let $g : \mathbb{R} \to \mathbb{R}$ be defined by $g(x) := x + 2x^2 \sin(1/x)$ for $x \neq 0$ and $g(0) := 0$. Show that $g'(0) = 1$, but in every neighborhood of 0 the derivative $g'(x)$ takes on both positive and negative values. Thus $g$ is not monotonic in any neighborhood of 0.

11. Give an example of a uniformly continuous function on [0,1] that is differentiable on (0, 1) but whose derivative is not bounded on (0, 1).

12. If $h(x) := 0$ for $x < 0$ and $h(x) := 1$ for $x \geq 0$, prove there does not exist a function $f : \mathbb{R} \to \mathbb{R}$ such that $f'(x) = h(x)$ for all $x \in \mathbb{R}$. Give examples of two functions, not differing by a constant, whose derivatives equal $h(x)$ for all $x \neq 0$.

13. Let $I$ be an interval and let $f : I \to \mathbb{R}$ be differentiable on $I$. Show that if $f'$ is positive on $I$, then $f$ is strictly increasing on $I$.

14. Let $I$ be an interval and let $f : I \to \mathbb{R}$ be differentiable on $I$. Show that if the derivative $f'$ is never 0 on $I$, then either $f'(x) > 0$ for all $x \in I$ or $f'(x) < 0$ for all $x \in I$.

15. Let $I$ be an interval. Prove that if $f$ is differentiable on $I$ and if the derivative $f'$ is bounded on $I$, then $f$ satisfies a Lipschitz condition on $I$. (See Definition 5.4.4.)